JustorAI

Bangladesh Legal Intelligence Platform

Problem & Fix Summary

Technical Diagnosis — Pilot Readiness Report

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8 Known Bugs	$10.02 Total Fix Cost	3-Week Timeline
accuracy failures	to pilot-ready	to Bar Council pilot

WHAT THIS DOCUMENT COVERS

01 Root causes of AI accuracy failures
02 All 8 known bugs with exact corrections
03 Embedding & LLM model upgrade plan
04 Cost breakdown (embeddings + API)
05 3-week pilot execution checklist

DIAGNOSIS

Why JustorAI gives wrong answers

The core product promise is Zero Hallucination — the AI answers only from verified Bangladeshi law. Currently that promise is broken on at least 8 confirmed cases. All three root causes below must be fixed before the Habiganj Bar Council pilot.

Root Cause 1 — Weak system prompt

The LLM's guardrails are not strict enough. When retrieved context is incomplete, the model fills the gap from its training memory — which is mostly Indian law (IPC, Indian CrPC, Indian CPC). The fix is a completely rewritten system prompt with hard-enforced citation rules: cite only retrieved sources, refuse if uncertain, never use training memory for legal claims.

Root Cause 2 — No section-number hard filter

When a user asks about Section 438 CrPC, the Supabase RPC returns 6 similar sections and lets the LLM pick. It picks the wrong one. The fix is a hard filter in match_acts_v2: if a section number is detected in the query, only return chunks from that exact section. This single change eliminates the most common category of wrong-section errors.

Root Cause 3 — 512 MB RAM ceiling on Render free tier

The current embedding model (all-MiniLM-L6-v2, 384-dim) was chosen because it fits in limited RAM. But it has no Bangla language understanding — users typing in Bangla or Banglish get poor retrieval results. Better multilingual models need 1.5–2.3 GB RAM, impossible on Render free tier. The fix is moving embeddings to the OpenRouter API (Qwen3-Embedding, 1024-dim, native Bengali support), freeing 80 MB RAM as a bonus.

THE 8 KNOWN BUGS

Exact errors in the current knowledge base

Every bug below is a case where the AI either used Indian law or returned a wrong number because the JSON data was incorrect or missing. Sanjib corrects these in the JSON files; Mehedi re-ingests after each fix.

Bug / Error	Wrong answer	Correct answer
Section 438 CrPC — anticipatory bail	Explains anticipatory bail as if 438 exists	438 was OMITTED from BD CrPC. Refer to Section 498
Limitation period — specific performance	Says limitation is 3 years	Correct answer: 1 year (Article 113, Limitation Act 1908)

Bug / Error	Wrong answer	Correct answer
Grandson inheritance under Muslim law	Sometimes says grandsons cannot inherit	MFLO Section 4 — Doctrine of Representation applies
Defamation — Section 500 arrest power	Implies police can arrest without warrant	Section 500 is non-cognizable — police CANNOT arrest without warrant
CPC Sections 100-103 — second appeal	Explains second appeal provisions from Indian CPC	These sections were OMITTED from Bangladesh CPC in 1978
Executive Magistrate powers — Section 29C	Confuses Executive and Judicial Magistrate powers	Executive Magistrates have limited powers; only CJM can award 7-yr imprisonment
Land ownership ceiling	Returns wrong bigha number (100, 500, or 'no limit')	Correct ceiling: 60 standard bighas per family (Land Reforms Ordinance 1984)
Wealth tax vs surcharge	References 'wealth tax rate' as if it exists	Bangladesh has NO wealth tax — it is a surcharge on net wealth (Income Tax Act 2023)

Additionally: all 33 law JSONs are missing the Source_Reference field, and all 10 DLR case files are missing DLR_Volume, DLR_Series, DLR_Citation fields. These must be added before re-ingestion.

MODEL UPGRADE

Embedding + LLM strategy for pilot

Embedding model — switch to Qwen3

The current model (all-MiniLM-L6-v2) has zero Bangla language support. Users asking in Bangla or Banglish get poor retrieval. The replacement (Qwen3-Embedding-0.6B via OpenRouter) natively supports Bengali, has 1024 dimensions vs 384, and ranks #1 on the MTEB multilingual leaderboard.

Property	all-MiniLM-L6-v2 (current)	Qwen3-Embedding-0.6B (new)
Dimensions	384	1024
Bangla support	None	Explicitly supported
Hosting	Self-hosted (80 MB RAM)	OpenRouter API (no RAM used)
Cost	$0 (but eats RAM)	~$0.02/M tokens
MTEB multilingual rank	Not ranked	#1 (8B size)
Re-ingestion cost (2,661 sections)	—	~$0.016 one-time

Why NOT nvidia/llama-nemotron-embed-vl-1b-v2:free?

It is a vision model — built for embedding scanned page images, not multilingual text. Its free endpoint explicitly states: 'All prompts and output are logged. Trial use only. Do not use for production or business-critical systems.' User legal queries cannot be logged to NVIDIA's training pipeline. Rejected.

LLM routing — three personas, three models

All three models run on Groq (fastest inference). OpenRouter serves as silent fallback when any Groq endpoint returns 429. One account each — no ToS violations.

Model	RPM	RPD	TPM	Notes
General Public Groq: llama-3.1-8b-instant	30	14,400	6,000	Fastest model. Highest daily limit. Plain-language format. Simple queries.
Law Student Groq: llama-4-scout-17b	30	1,000	30,000	17B MoE model. 128K context. Highest TPM on free tier — handles long law sections.
Legal Professional Groq: qwen3-32b	60	1,000	6,000	Largest free model. 60 RPM (2× others). Best for IRAC legal reasoning.
Fallback (all) OpenRouter: same models :free	20	1,000*	—	*1,000/day requires $10 credit top-up. Without it: only 50/day total.

Pilot daily capacity: ~250 general public queries, ~1,000 law student requests, ~200 lawyer IRAC analyses. More than sufficient for 20–50 pilot users.

COST BREAKDOWN

Total bootstrapped spend for pilot readiness

Scenario	Queries / day	Monthly tokens	Monthly embed cost
Pilot (conservative)	100	300,000	**$0.006 / month**
Pilot (active)	500	2,250,000	**$0.045 / month**
Post-pilot growth	2,000	9,000,000	**$0.18 / month**
10K users/day	10,000	45,000,000	**$0.90 / month**

Embedding cost at $0.02/M tokens (Qwen3-Embedding-0.6B via OpenRouter). One-time ingestion of all 33 law files: ~$0.016.

Item	Cost	When
Qwen3 re-ingestion (2,661 law sections)	$0.016	Week 1, once
OpenRouter credit top-up (unlocks 1,000 RPD fallback)	$10.00	Before pilot
Groq LLM calls (all three personas)	$0.00	Free tier
Render paid tier (no cold starts)	$7.00/mo	Week 2
Total API cost to pilot-ready state	**$10.02**	—

EXECUTION PLAN

3-week checklist to pilot-ready

Mehedi (technical) and Sanjib (legal data) work in parallel during Week 1. Both tracks must complete before Week 2 testing begins. Pilot cannot proceed until the golden test suite scores 90%+.

Week	Owner	Tasks
Week 1 Mehedi	Infrastructure (Technical)	• Sign up OpenRouter, create 3 API keys (embedding / lawstudent / lawyer) • Add $10 credit to OpenRouter (unlocks 1,000 RPD fallback) • Update Supabase schema: vector dim 384 → 1024, add DLR citation columns • Replace embedding in backend.py with OpenRouter Qwen3-Embedding-0.6B API • Remove sentence-transformers + torch from requirements.txt (saves 80 MB RAM) • Update match_acts_v2 RPC: add section-number hard filter • Rewrite system prompts: 3 personas (General Public / Law Student / Lawyer) • Implement Groq → OpenRouter fallback on 429 • Deploy to Render, verify /health endpoint • Run validate_json.py — send report to Sanjib
Week 1 Sanjib	Data Fixes (Legal)	• Fix all 8 bugs in JSON files (exact corrections documented in Bug.md) • Add Source_Reference field to all 33 law JSON files • Add DLR_Volume, DLR_Series, DLR_Citation to all 10 DLR case files • Manual spot-check: all omitted/repealed sections, all numeric values • Run AI cross-check prompt on any new law PDFs converted this week • Review all flagged issues from validate_json.py report
Week 2 Mehedi	Testing (Technical)	• Run ingest_v2.py --replace (full re-ingestion at 1024-dim) • Run test_golden_queries.py — target 90%+ accuracy score • For every failing test: Sanjib corrects JSON → Mehedi re-ingests → retest • Add law dictionary as static JSON lookup endpoint (GET /dictionary?term=...) • Upgrade Render to paid tier ($7/mo) — eliminates cold-start delays • Add daily query limit guard (50/day pilot cap per user) • Final test run — must score 90%+ before proceeding
Week 3 All	Pilot (Go-Live)	• Demo to Habiganj Bar Council • Collect structured feedback per query (which persona, which law, pass/fail) • Track: which queries still fail, what lawyers say about IRAC format • Prioritise next 20 DLR cases to add based on lawyer feedback • Document accuracy score before and after pilot feedback cycle

Golden Test Suite — 17 test cases (test_golden_queries.py)

Each test specifies must_contain phrases (e.g. 'omitted', '1 year', 'non-cognizable') and must_not_contain phrases (e.g. 'anticipatory bail under 438', 'Indian Penal Code'). A response passes only if all must_contain phrases are present and no must_not_contain phrase appears. Score = passed / 17. Target: 15/17 (88%) minimum, 17/17 (100%) goal.

QUICK REFERENCE

Key model IDs and API endpoints

Purpose	API	Model ID	Endpoint
Embedding (query + ingest)	OpenRouter	qwen/qwen3-embedding-0.6b	openrouter.ai/api/v1/embeddings
General Public LLM	Groq	llama-3.1-8b-instant	api.groq.com/openai/v1/chat/completions
Law Student LLM	Groq	meta-llama/llama-4-scout-17b-16e-instruct	api.groq.com/openai/v1/chat/completions
Legal Professional LLM	Groq	qwen/qwen3-32b	api.groq.com/openai/v1/chat/completions
Fallback (all personas)	OpenRouter	Same model IDs with :free suffix	openrouter.ai/api/v1/chat/completions

One sentence summary for leadership:

JustorAI's AI has the right law sections in its database — the problem is the model ignores them and answers from Indian law training memory instead. Fixing the system prompt, adding a section filter, and upgrading the embedding model costs $10.02 total and can be completed in 3 weeks.

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